1. Introduction

1.1. Background and Motivation

Autonomous and cyber–physical systems increasingly operate in environments where their performance depends on the decisions of another agent. In these settings, the control problem is not only to stabilize a plant or track a prescribed reference; rather, the controller must act while another decision maker attempts to influence the same state in an opposing direction. Pursuit–evasion games provide one of the clearest mathematical representations of this interaction. A pursuer attempts to reduce a relative error or drive the state toward a capture condition, while an evader attempts to increase the same cost or prevent the capture condition from being reached. As a result, the pursuit–evasion problem is able to convert an open-ended adversarial interaction into a controlled two-player game whose state, inputs, objective, and equilibrium conditions are explicitly defined [1, 11].

The linear-quadratic specialization is especially useful because it retains the strategic structure of the game while allowing the equilibrium policies to be computed from matrix equations. More specifically, a discrete-time linear system with quadratic state and input penalties leads to a two-player zero-sum game in which one player minimizes the performance index while the other maximizes it. The corresponding saddle-point solution is connected to a game algebraic Riccati equation (GARE), and the resulting policies are linear functions of the state. This formulation is important since it gives a transparent model-based baseline whose assumptions, numerical conditions, and closed-loop behavior can be examined directly [1, 2].

The model-based solution, however, is only one side of the control problem. A GARE-based controller requires the system matrices to be known, while many practical systems provide measured input-output or state-transition data without providing an exact analytical model. Model-free quadratic Q-learning addresses this difference by estimating an action-value function from sampled data and recovering the same game policies from the estimated quadratic matrix. Early work demonstrated that a linear discrete-time zero-sum game could be solved by model-free Q-learning and that the resulting iterations converge to the nominal Riccati solution under the required excitation and rank conditions [3, 4]. Later policy-iteration Q-learning methods further developed this idea for data-based two-player zero-sum systems [5].

This model-based versus model-free distinction should not be interpreted as a comparison between an old analytical method and an unrelated learning method. Both approaches solve the same zero-sum game, use the same quadratic objective, and seek the same pursuer and evader policies. Their principal difference is the information used during the computation. The model-based methods evaluate or update the value matrix using the known system dynamics, whereas the model-free methods reconstruct the required quadratic relationships from measured transitions. Rizvi and Lin organize these ideas through corresponding policy-iteration and value-iteration formulations for nominal linear quadratic zero-sum games [6]. This shared structure creates a direct foundation for studying four algorithms under the same problem definition: model-based policy iteration, model-based value iteration, model-free Q-learning policy iteration, and model-free Q-learning value iteration.

Recent research continues to refine each side of this framework. New model-based work has developed single-loop policy-iteration methods for linear zero-sum games with relaxed initialization requirements [7]. Recent value-iteration research has also established on-policy and off-policy formulations for stochastic zero-sum linear quadratic games with unknown dynamics [8]. In parallel, modern learning theory has investigated convergence, last-iterate behavior, and sample complexity for zero-sum linear quadratic games [9]. These developments demonstrate that the linear quadratic zero-sum game remains an active research setting rather than only a historical control benchmark.

Nevertheless, much of the nominal algorithmic literature assumes that the input computed by the controller is the input that enters the plant. This assumption is convenient for the derivation, but it excludes a common execution-side problem: a persistent difference between a commanded and realized input. Such a difference may represent an actuator calibration offset, steering-center shift, longitudinal-command mismatch, sensor-to-command conversion error, or another repeatable implementation bias. Even when the nominal feedback gains stabilize the game, a persistent input offset introduces an affine forcing term into the closed-loop dynamics and may create a nonzero deviation from the nominal trajectory.

The distinction is crucial because nominal convergence of the value matrix does not imply that the realized system will follow the nominal closed-loop response. A policy may satisfy the GARE, the saddle-point curvature conditions, and the closed-loop stability condition while the implemented trajectory still settles away from its nominal equilibrium. Recent fault-tolerant Q-learning studies further demonstrate the current interest in learning-based control under actuator and sensor faults rather than only under ideal input channels [10]. In pursuit–evasion research, current work has similarly moved toward safety, disturbance rejection, sensor limitations, and physical implementation [12, 13, 14].

Therefore, the direction of this thesis is to preserve the established four-algorithm zero-sum framework and extend it with a common persistent command-offset model, bias-estimation procedure, affine compensation law, and MATLAB validation structure. The contribution is not the invention of the underlying policy-iteration, value-iteration, or Q-learning equations. Rather, the contribution is the systematic development of disturbance-aware versions that remain traceable to the original nominal algorithms while allowing the effect of imperfect command realization to be measured and compensated.

1.2. Problem Statement

This thesis considers a discrete-time two-player zero-sum system in which the pursuer acts through one input channel and the evader acts through another. The nominal game dynamics are represented by

$$x_{k+1} = Ax_k + B_P u_{P,k} + B_E u_{E,k},$$

where $x_k \in \mathbb{R}^n$ denotes the game state, $u_{P,k} \in \mathbb{R}^{m_P}$ denotes the pursuer input, $u_{E,k} \in \mathbb{R}^{m_E}$ denotes the evader input, and A, B_P, and B_E are the state and input matrices. The pursuer is the minimizing player, while the evader is the maximizing player.

A representative infinite-horizon quadratic objective is

$$J = \sum_{k=0}^{\infty} \left(x_k^\top Q x_k + u_{P,k}^\top R_P u_{P,k} - u_{E,k}^\top R_E u_{E,k} \right),$$

where $Q \succeq 0$, $R_P \succ 0$, and $R_E \succ 0$. Under the nominal saddle-point policies, the two inputs are written as

$$u_{P,k} = Lx_k, \qquad u_{E,k} = Kx_k,$$

where L and K are the pursuer and evader feedback gains. The corresponding closed-loop matrix is $A + B_P L + B_E K$.

The nominal model-based methods obtain the value matrix and the two gains from GARE-based policy-iteration or value-iteration updates. The nominal model-free methods estimate a quadratic Q-function from measured data and recover the same policies from the partitions of the estimated action-value matrix. In the algorithms used as the starting point for this thesis, these two pairs of methods have different information requirements, but they are directed toward the same saddle-point solution [6].

The practical difficulty considered here is that the realized player inputs may differ from their commanded values. The command-channel model is given by

$$u^{\mathrm{act}}_{P,k} = u^{\mathrm{cmd}}_{P,k} + b_P, \qquad u^{\mathrm{act}}_{E,k} = u^{\mathrm{cmd}}_{E,k} + b_E,$$

where b_P and b_E are constant or slowly varying offsets over the calibration interval and the following closed-loop run. When no compensation is applied, the realized dynamics become

$$x_{k+1} = (A + B_P L + B_E K)x_k + B_P b_P + B_E b_E.$$

As a result, the nominal gains may remain stabilizing while the affine offset changes the steady-state response, the state trajectory, and the accumulated game cost.

The command offsets are estimated according to the information available to each algorithmic class. For the model-based methods, the known system matrices allow a transition residual to be formed from measured states and commanded inputs. For the model-free methods, the offset estimate is obtained from measured command-channel data or from an augmented data regression that does not require the analytical state-transition matrices in the Q-learning update.

The compensated commands are then written as

$$u_{P,k}^{\text{cmd}} = Lx_k - \hat{b}_P, \qquad u_{E,k}^{\text{cmd}} = Kx_k - \hat{b}_E.$$

Therefore, the compensated closed-loop system becomes

$$x_{k+1} = (A + B_P L + B_E K)x_k + B_P(b_P - \hat{b}_P) + B_E(b_E - \hat{b}_E).$$

This equation gives the central structure of the thesis. If the offset estimates are exact, the nominal closed-loop dynamics are recovered. If the estimates are imperfect, the remaining affine term depends only on the estimation errors rather than on the full command offsets.

The research problem is consequently not to replace the original GARE or Q-learning algorithms. It is to determine whether the four established methods can be extended under a common disturbance model while their original nominal iteration structures remain identifiable. The extended implementations must also be evaluated using common numerical criteria so that differences caused by the information structure are not confused with differences caused by inconsistent simulation conditions.

1.3. Research Questions and Objectives

This thesis is structured around five research questions that connect the nominal zero-sum solution, the command-offset model, the estimation process, the compensation law, and the MATLAB comparison.

1) *Nominal Algorithm Reproduction:* Can the model-based policy-iteration, model-based value-iteration, model-free Q-learning policy-iteration, and model-free Q-learning value-iteration methods be implemented so that their nominal equations remain traceable to the established zero-sum algorithms from which they are derived?

2) *Persistent Command-Offset Effect:* How do constant pursuer and evader command offsets alter the nominal state response, predicted steady-state deviation, finite-window game cost, and closed-loop trajectory when the feedback gains themselves are unchanged?

3) *Bias Estimation and Compensation:* Can the persistent command offsets be estimated using the information available to the model-based and model-free methods, and can affine input compensation reduce the resulting deviation from nominal behavior?

4) *Matched Four-Algorithm Comparison:* How do policy iteration and value iteration compare within the model-based and model-free classes when all four methods use the same game matrices, disturbance definitions, calibration conditions, simulation horizon, and numerical diagnostics?

5) *Pursuit–Evasion and Future Implementation Relevance:* To what extent does the resulting disturbance-aware framework provide a defensible mathematical and computational foundation for later pursuit–evasion studies involving nonlinear vehicles, QLabs, or physical QCar platforms?

From these research questions, the thesis defines five primary objectives.

- *Objective 1:* Reproduce and document the nominal model-based and model-free policy-iteration and value-iteration methods without presenting the underlying algorithms as new contributions.
- *Objective 2:* Introduce a common persistent command-channel offset model for the pursuer and evader and derive its effect on the realized closed-loop dynamics.
- *Objective 3:* Develop information-consistent offset-estimation and affine-compensation procedures for the model-based and model-free implementations.
- *Objective 4:* Establish a shared MATLAB validation protocol using convergence, GARE residual, closed-loop spectral radius, saddle-point curvature, regression rank, conditioning, bias-estimation error, trajectory deviation, steady-state offset, and finite-window cost.
- *Objective 5:* Interpret the resulting zero-sum system as a pursuit–evasion benchmark and identify the precise work that remains before nonlinear vehicle or hardware claims can be made.

1.4. Scope, Assumptions, and Delimitations

There are several scope decisions that are intentionally restrictive. First, the thesis considers one pursuer and one evader. Multi-pursuer and multi-evader games are important within the broader pursuit–evasion literature; however, they introduce coordination, assignment, communication, and scalability questions that would make it difficult to isolate the disturbance-aware extension of the four algorithms. The two-player case is therefore used as the controlled setting in which the model-based and model-free information structures can be compared directly.

Second, the primary algorithmic development is based on discrete-time linear systems and quadratic zero-sum objectives. The linear formulation is not claimed to represent every physical pursuit–evasion system globally. Rather, it provides the exact setting in which the GARE, the quadratic Q-function, and the pursuer and evader feedback gains can be compared without ambiguity. Locally linearized nonlinear models may later be connected to this framework, but the mathematical claims in the present thesis are restricted to the linear or locally linearized problem used by each experiment.

Third, the command disturbances are modeled as constant or slowly varying offsets over a calibration batch and the following closed-loop run. This assumption is intended to represent repeatable execution-side mismatch, not arbitrary high-frequency noise. Time-varying faults, abrupt actuator loss, stochastic drift, and unknown nonlinear dead zones are outside the principal scope. These effects are relevant future extensions, but they require estimators and robustness arguments that differ from the persistent affine model considered here.

Fourth, the model-based methods are allowed to use the analytical system matrices during policy or value computation and during transition-residual calibration. The model-free Q-learning updates are not provided with the analytical state-transition matrices. This distinction is maintained throughout the thesis. At the same time, the model-free designation does not prohibit the use of measured commanded and realized inputs, since those quantities may be available from the implementation interface even when the state-space matrices are unknown.

Fifth, the thesis focuses on structured quadratic Q-learning rather than a broad reinforcement-learning algorithm comparison. Deep actor-critic methods, self-play, neural policies, and large-scale multi-agent training are outside the main scope. Recent pursuit–evasion research demonstrates the usefulness of those methods under nonlinear dynamics and sensing limitations [12, 14]; however, introducing them here would obscure the direct relationship between the learned Q-function and the nominal GARE solution.

Sixth, MATLAB serves as the primary validation environment. The thesis reports numerical convergence, stability, estimation, compensation, and trajectory results. It does not claim completed QLabs deployment or physical QCar validation. The autonomous-ground-vehicle interpretation remains important as an application pathway, but the current work is theoretical and computational in scope.

Finally, the thesis does not claim that policy iteration, value iteration, GARE theory, or model-free Q-learning were invented in this work. The nominal algorithmic equations are established in the cited literature. The scope of the contribution is the disturbance-aware extension, the matched implementation across four methods, and the resulting numerical evidence.

1.5. Contributions and Novelty

The first contribution is a common disturbance-aware formulation for model-based and model-free zero-sum control. The same pursuer and evader command-offset model is introduced into all four methods, which prevents the comparison from being divided into unrelated experiments. This common formulation makes it possible to identify which differences arise from policy iteration versus value iteration and which differences arise from model knowledge versus sampled-data learning.

The second contribution is the extension of the model-based GARE policy-iteration and value-iteration implementations with persistent-offset calibration and affine compensation. The nominal policy-evaluation, policy-improvement, and value-update structures remain the established model-based algorithms. The additional components estimate the command offsets, apply compensation during execution, and quantify the residual deviation when the estimates are imperfect.

The third contribution is the corresponding extension of the model-free quadratic Q-learning policy-iteration and value-iteration implementations. The Q-learning methods retain the sampled Bellman regression and gain-recovery structure of the nominal algorithms. The disturbance-aware implementation adds commanded-versus-realized input accounting, offset estimation under the model-free information structure, and compensated closed-loop simulation.

The fourth contribution is a shared numerical validation protocol. Each method is evaluated using nominal operation, persistent offsets without compensation, persistent offsets with estimated compensation, and exact compensation as a verification case. The reported diagnostics distinguish algorithm convergence from game validity and execution performance. More specifically, the MATLAB studies include the value or Q-function iteration history, gain convergence, GARE or Bellman residuals, closed-loop spectral radius, saddle-point curvature, matrix conditioning, feature rank, bias-estimation error, trajectory deviation, predicted steady-state offset, and finite-window game cost.

The fifth contribution is the explicit separation between nominal algorithm validity and disturbance-compensation performance. This separation is necessary because a controller can satisfy the nominal GARE while the realized system experiences an affine input error. By reporting both the nominal numerical checks and the disturbance-aware trajectory results, the thesis avoids using a single convergence plot as evidence for every aspect of the implementation.

The novelty of the thesis therefore lies in the integration and matched evaluation of these components. It does not lie in renaming the established algorithms or presenting existing Riccati and Q-learning updates as original. This distinction strengthens the thesis because the reader is able to identify precisely where the literature ends and where the disturbance-aware development begins.

1.6. Connections to the Literature

There are six closely connected bodies of literature that support this thesis. The first is linear quadratic zero-sum dynamic game theory. This literature establishes the minimax problem, the quadratic value function, the feedback saddle-point policies, and the GARE conditions used by the model-based solution [1, 2]. It also provides the theoretical connection between a controller acting as the minimizing player and an adversarial disturbance or evader acting as the maximizing player.

The second body of literature concerns model-based policy iteration and value iteration for zero-sum games. The classical iterative structure is developed from policy evaluation, policy improvement, and repeated Riccati mappings. Recent work continues to study initialization, single-loop implementation, and convergence of GARE iterations [7]. These results support the two model-based algorithms while also showing that their numerical implementation remains an active topic.

The third body of literature concerns quadratic Q-learning and adaptive dynamic programming. Foundational discrete-time work established that the zero-sum game policies could be learned without direct knowledge of the system matrices [3]. Output-feedback and data-based developments then expanded the available information structures and demonstrated policy-iteration learning from measured data [4, 5]. The monograph by Rizvi and Lin provides the direct organization of policy-iteration and value-iteration methods used as the nominal algorithmic foundation for this thesis [6].

The fourth body of literature addresses current learning and optimization results for zero-sum linear quadratic games. Recent research has examined on-policy and off-policy value iteration with unknown dynamics, as well as sample complexity and last-iterate convergence for model-free learning [8, 9]. This literature is important because it positions the thesis within the present development of structured zero-sum learning rather than within a generic reinforcement-learning comparison.

The fifth body of literature concerns actuator faults, persistent input mismatch, estimation, and compensation. Fault-tolerant Q-learning methods increasingly attempt to preserve stability and optimality when the realized system contains actuator or sensor faults [10]. The present thesis uses a narrower constant-offset model, but this literature supports the larger motivation that learning and control algorithms must be evaluated under imperfect execution channels rather than only under nominal equations.

The sixth body of literature concerns pursuit–evasion games and their modern applications. Pursuit–evasion provides the adversarial interpretation of the two-player zero-sum system [11]. Recent surveys show that learning-based pursuit–evasion research now spans unmanned systems, navigation, and spacecraft applications [12]. Current studies also address safety, stability, disturbances, sensing limits, and car-like dynamics [13, 14]. These sources provide the application context while the thesis remains focused on a controlled linear quadratic foundation.

1.7. *Significance of the Thesis*

The significance of this thesis is methodological as well as practical. Methodologically, it creates a direct comparison among four algorithms that are often discussed separately. Model-based policy iteration, model-based value iteration, Q-learning policy iteration, and Q-learning value iteration are placed under the same system, cost, disturbance, and reporting structure. As a result, the comparison does not depend on changing the game definition from one algorithm to another.

The work is also significant because it treats command realization as part of the control problem. In many numerical studies, the commanded input is inserted directly into the state equation. This convention is useful for analyzing the nominal algorithm but can hide a repeatable implementation mismatch. A persistent offset does not need to destabilize the system in order to matter. It may instead produce a nonzero state deviation, change the game cost, or alter the interpretation of the pursuer and evader policies. The compensation framework makes this effect explicit.

Another point of significance is the separation of information structures. The model-based methods use known dynamics, while the model-free methods recover the game policies from data. Applying the same command-offset experiment to both classes provides a more meaningful comparison than simply comparing their nominal iteration counts. It reveals the additional questions introduced by data collection, feature rank, regression conditioning, and offset measurement.

The thesis is also intended to be reproducible. Each MATLAB implementation records its parameters, convergence histories, numerical checks, calibration results, trajectories, and summary metrics. This reproducibility is important because it allows later work to replace the benchmark matrices with a vehicle model without changing the logic of the experiment. More specifically, scalar command offsets can later become acceleration and steering offset vectors, while the nominal, uncompensated, estimated-compensation, and exact-compensation cases remain unchanged.

Finally, the thesis provides a defensible intermediate contribution. It is more complete than a nominal GARE example because it includes disturbance estimation and compensation across model-based and model-free methods. At the same time, it is more controlled than a full nonlinear vehicle or deep multi-agent learning study. This intermediate position is useful because it clarifies which results are established in MATLAB and which results still require nonlinear simulation, QLabs integration, or physical hardware testing.

1.8. Working Hypotheses

Although this thesis is comparative rather than purely confirmatory, several working hypotheses guide the algorithm development and numerical experiments.

1) *Nominal Consistency Hypothesis:* When the four algorithms are implemented under compatible assumptions and sufficient numerical conditions, the model-based and model-free methods will approach feedback policies associated with the same nominal zero-sum solution.

2) *Persistent-Offset Hypothesis:* Constant pursuer and evader command offsets will introduce a measurable affine deviation from the nominal trajectory even when the nominal closed-loop matrix remains Schur stable.

3) *Exact-Compensation Hypothesis:* When the true command offsets are subtracted from the commanded inputs, the realized trajectory will recover the nominal closed-loop trajectory to numerical precision.

4) *Estimated-Compensation Hypothesis:* When the offsets are estimated with small error, affine compensation will reduce the steady-state and trajectory deviation by an amount consistent with the residual bias terms $b_P - \hat{b}_P$ and $b_E - \hat{b}_E$.

5) *Information-Structure Hypothesis:* The model-free implementations will require stronger data-rank, excitation, and conditioning diagnostics than the model-based implementations because the quadratic game relationships must be reconstructed from sampled transitions.

6) *Transfer-Foundation Hypothesis:* A disturbance-aware framework that is verified first on a controlled zero-sum linear system will provide a more defensible foundation for later vehicle experiments than transferring a nominal controller directly to a platform without an explicit command-mismatch model.

These hypotheses do not predetermine that one algorithmic class must outperform the other.

A model-based method may converge more directly because the matrices are known, while a model-free method may remain valuable because it does not require those matrices in the learning update. The purpose of the experiments is to document these differences under matched conditions rather than to force a predetermined ranking.

1.9. Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews the literature on zero-sum linear quadratic games, GARE policy iteration, GARE value iteration, quadratic Q-learning, persistent input bias, disturbance estimation, affine compensation, numerical well-posedness, and current pursuit–evasion control. The chapter separates foundational algorithm sources from recent state-of-the-art developments so that the historical basis and the current research direction remain clear.

Chapter 3 develops the technical background and common disturbance-aware game formulation. It defines the discrete-time two-player zero-sum system, quadratic cost, nominal saddle-point policies, GARE structure, joint quadratic Q-function, persistent command-channel offsets, estimation assumptions, compensation law, and common numerical criteria.

Chapter 4 presents the model-based methods. The original policy-iteration and value-iteration structures are documented first, after which the persistent-offset calibration, affine compensation, MATLAB simulation cases, and numerical diagnostics are added. The chapter distinguishes changes to the experiment from changes to the underlying algorithms.

Chapter 5 presents the model-free methods. It develops the quadratic Q-function representation, sampled Bellman equations, feature construction, least-squares regression, policy recovery, and the corresponding policy-iteration and value-iteration procedures. The same command-offset model and compensation cases are then applied under the model-free information structure.

Chapter 6 reports the MATLAB results and cross-method comparison. It evaluates convergence, game validity, closed-loop stability, regression quality, bias estimation, uncompensated behavior, estimated compensation, exact compensation, and finite-window game cost. The chapter uses matched tables and figures so that the four algorithms can be compared without changing the benchmark definition.

Chapter 7 concludes the thesis by summarizing the main findings, answering the research questions, identifying limitations, and defining future work. The future directions include nonlinear kinematic bicycle models, vector-valued acceleration and steering offsets, online bias estimation, output-feedback implementation, actuator saturation, time delay, QLabs simulation, physical QCar testing, and multi-pursuer or multi-evader extensions.

References Cited in Chapter 1

[1] T. Basar and G. J. Olsder, *Dynamic Noncooperative Game Theory*, 2nd ed. Philadelphia, PA: SIAM, 1999.

[2] M. Pachter and K. D. Pham, "Discrete-time linear-quadratic dynamic games," *Journal of Optimization Theory and Applications*, vol. 146, pp. 151–179, 2010.

[3] A. Al-Tamimi, F. L. Lewis, and M. Abu-Khalaf, "Model-free Q-learning designs for linear discrete-time zero-sum games with application to H_∞ control," *Automatica*, vol. 43, no. 3, pp. 473–481, 2007.

[4] S. A. A. Rizvi and Z. Lin, "Output feedback Q-learning for discrete-time linear zero-sum games with application to the H_∞ control," *Automatica*, vol. 95, pp. 213–221, 2018.

[5] B. Luo, Y. Yang, and D. Liu, "Policy iteration Q-learning for data-based two-player zero-sum game of linear discrete-time systems," *IEEE Transactions on Cybernetics*, vol. 51, no. 7, pp. 3630–3640, 2021.

[6] S. A. A. Rizvi and Z. Lin, *Output Feedback Reinforcement Learning Control for Linear Systems*. Cham, Switzerland: Birkhäuser, 2023.

[7] J. Zhao, C. Yang, W. Gao, and J. H. Park, "Novel single-loop policy iteration for linear zero-sum games," *Automatica*, vol. 163, Art. no. 111551, 2024.

[8] L. Guo, B.-C. Wang, and J.-F. Zhang, "On-policy and off-policy value iteration algorithms for stochastic zero-sum dynamic games," *Journal of Systems Science and Complexity*, vol. 38, pp. 421–435, 2025.

[9] J. Wu, A. Barakat, I. Fatkhullin, and N. He, "Learning zero-sum linear quadratic games with improved sample complexity and last-iterate convergence," *SIAM Journal on Control and Optimization*, vol. 63, no. 5, pp. 3244–3271, 2025.

[10] M. Kankashvar, S. Rafiee, and H. Bolandi, "Fault-tolerant Q-learning for discrete-time linear systems with actuator and sensor faults using input-output measured data," *Franklin Open*, Art. no. 100259, 2025.

[11] I. E. Weintraub, M. Pachter, and E. Garcia, "An introduction to pursuit-evasion differential games," in *Proceedings of the American Control Conference*, 2020.

[12] K. Yang, A. Shen, N. Xu, F. Deng, M. Lu, and C. Chen, "A review of reinforcement learning approaches for pursuit-evasion games," *Chinese Journal of Aeronautics*, 2025, Art. no. 103940.

[13] X. Wang, H. Zhang, J. Xu, S. Wang, and M. Guay, "Reinforcement learning in pursuit-evasion differential game: Safety, stability and robustness," arXiv:2507.19516, 2025.

[14] B. M. Gonultas and V. Isler, "Learning to play pursuit-evasion with dynamic and sensor constraints," arXiv:2405.05372, 2024.